A Code for the Refusal, a Record for the Success

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Draft, 13 November 2026. Every quotation is from a document opened on a primary; the sources, with routes, dates opened and the character ranges actually read, are in §9. **Not legal advice and not a legal reading — the instruments are quoted and the inferences are the author's. I am not a lawyer.**

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1. The question paper no. 5 left open, and the one sentence this paper defends

Paper no. 5 of this series read the liability machinery against a case where a software delegate spends inside its limits and outside its instructions: Part 7 of the Payment Services Regulations 2017, the common-law ledger as the courts have so far described it, and Regulation E. It found that the record which would settle such a case has three parts — the grant, the close, and the join between them — and its abstract then stops, deliberately, at the edge of the next question. The record, it says, *"is one nothing examined requires anyone to keep. What kind of instrument could require it is a question about instruments, and is not answered here."*

This paper reads the instruments. Eleven of them, in the order a transaction meets them, and eight authors who reached part of the same argument before this paper did, in two literatures that do not cite each other.

What it finds is not an absence. Refusals are required to be recorded, under MUSTs, by instruments and drafts counted in §§3 and 4; in one programme they are counted monthly at network scale with the arithmetic published. An argument built on the industry having no vocabulary for refusal would be wrong, and an earlier draft of this one was. What is left after the instruments are read is narrower, harder to fix, and the whole of what is claimed here:

> ***A refusal-inclusive denominator exists. In none of the instruments read does there exist one that someone other than its author can establish is complete.***

Section 2 is the sharpest single piece of evidence for it, because it comes from a publisher that computes both shapes in one programme. Sections 3 and 4 are the instruments and the eight authors. Section 5 states what survives and at what width. **Four claims an earlier draft of this argument made are withdrawn; §7 names them and says what broke each one.**

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2. The asymmetry, and one programme computes both halves of it

The claim this paper is about can be put as arithmetic. **Rates about authorisation are computed over the transactions that completed.** The refusals — the attempt that was declined, the mandate that was refused, the instruction that never entered — are in the numerator of nothing and the denominator of nothing, so a reader cannot tell whether a conformance figure of 0.97 describes a careful system or a system whose failures were never written down.

An earlier draft argued this from a field-presence rule in a schema. That evidence stands and is in §3. But there is better, and it comes from the one place that had no reason to make the paper's point: **a production monitoring programme that computes two rates, one of each shape, and publishes both definitions.**

The Visa Acquirer Monitoring Program's fact sheet gives its principal rate as

> ***VAMP Ratio = Count of [Fraud (TC40) + Disputes (TC15)] ÷ Count of Settled Transactions (TC05)***

and its enumeration rate as

> *numerator* ***"[Count of Enumerated Authorization Transactions (Approved + Declined)]"****,*
> *denominator* ***"[Count of Authorization Transactions (Approved+ Declined)]"***

Both are computed by Visa, monthly, over VisaNet: *"Each month, Visa monitors fraud, dispute, and enumeration levels and identifies acquirers or merchants that exceed monthly VAMP thresholds."*

The denominators are different and the difference is not an oversight. Enumeration — carding, the systematic testing of card numbers — is visible in declines. A rate computed over settlements would measure nothing, because the attack's signature is the attempts that failed. So the declines are in the denominator, explicitly, in both terms. The fraud-and-dispute rate, which is where an acquirer's or a merchant's conduct is assessed, is computed over settled transactions, and what a merchant refused is not in it.

One publisher, one programme, two ratios, and **the declines are in the denominator of the rate whose subject is an attack visible only in declines, and out of the rate whose subject is a merchant's or an acquirer's conduct.** The fact sheet states both definitions and gives no reason for the difference, so nothing here is a claim about anyone's motive — the placement is the finding, and §4 shows the same placement in eight documents that had no connection to this one.

A standards draft from another field puts its requirement in the matching place. `draft-schrock-action-evidence-boundary` makes the refusal record a **MUST** at exactly one point — the pre-entry recheck, where the attempt is about to occupy an action key. §5.10: an explicit not-entered marker, bound to the attempt, *"in the same atomic write."* Where a verdict is merely formed, at the evaluator, the draft **SHOULD** emit a signed record carrying *"the separate AUTHORIZED or REFUSED verdict, as applicable"* — and nothing in the portion read says an attempt refused at that stage is an attempt *closed as not entered*, so the MUST does not reach it. The draft gives a reason for the SHOULD — *"so that this case cannot arise"* — and none for the MUST beyond atomicity with the close.

Two instruments, two fields, the same shape of difference. Whether anything follows from that is §4's question, and it is answered by counting rather than by reasoning about why.

3. The instruments, and what each one reaches

Eleven instruments, in the order a transaction meets them. Each entry says what it reaches and stops. **One of them is a statute whose relevant sections date from 1989**, which matters to how the pattern should be read: it is not a property of agentic protocols. *No claim is made about which instrument is*

oldest: the Nacha row below carries no date and was not read on the rule text.

The Agent Payments Protocol binds the receipt to the mandate by hash, as a required field, and then makes the two fields closest to identifying the movement — `psp_confirmation_id` and `network_confirmation_id` — **present only if status is Success**, by a field-presence rule in the schema. No refusal population is specified, no logging duty, no retention duty. *It reaches the join, for the transactions that succeeded.*

x402 settles on a public chain: the facilitator *"submits the payment to the blockchain based on the scheme and network"* and *"waits for the payment to be confirmed on the blockchain."* On the prioritised EIP-3009 path the settlement emits an event carrying an authorisation nonce, which a stranger can enumerate without asking anyone — and the path matters, because the specification says *"the implementation should prioritize `eip3009` (if compatible) and then `permit2`"* and scopes the Permit2 proxy to *"Tokens without EIP-3009"*, so the four-field event an earlier reading took for the core path is the fallback. *It reaches a stranger-readable record of the movement.* It specifies no refusal

population, no logging duty and no retention duty.

An Internet-Draft takes that handle and specifies the join, then declines to verify it. `draft-toraman-noa-action-digest`, read in full across both revisions on two publishers, puts the authorisation identifier in *"the external system's caller-chosen opaque identifier field"* under its mode (a). The obligation that would make the result checkable by a stranger is **E1**, and it falls on the external system: the field MUST be committed *"in a way no party in the dispatch path can alter after the embedding party signs."* The draft then disclaims its own enforcement of it — *"E1 constrains a third-party external system's behaviour, so no conformance vector in this corpus can enforce it — it is a residual on the deployment, not a control this document verifies"* — and says the same of E2, *"not vector-enforceable"*, and of E3, *"an obligation on the embedding party and its deployment, not a property any conformance vector can check."* **There is no at-most-once check.** The verification section's fifth step is a comparison, *"compare the recomputed digest to the value observed in the external system"*, and the section closes by refusing the inference such a comparison invites: *"A Relying Party MUST NOT conclude from a successful*

comparison that the action was performed, that any effect occurred, or that any statement about the action made by any party is true." It goes further about its own third step, whose construction *"verifies no signature"*, so a forged pair is *"ACCEPTED by step 3 and REFUSED by the record's own chain verification"* and *"A green step 3 over unverified documents establishes nothing." It reaches a specified join over a stranger-readable movement, with the part that would let an outsider check the result named three times, in its own normative text, as a residual it does not verify.*

An extension to x402 supplies the refusal code x402 itself does not, and it comes closest in this list to the paper's own title. `draft-vauban-x402-consolidated-00` (F. Serafini, Ed., Vauban Research, 28 May 2026; Informational, an Independent Submission and not a Working Group product) defines at §3.13 a set of seven reject tokens — `MalformedClaim`, `PaymentRequired`, `UnsupportedReceiptFormat`, `NullifierReplay`, `Expired`, `StructuralInvalid`, `HumanityRequired` — and closes it: *"The token set is closed for this version; new tokens require registration alongside the receipt-format registry."* The reason is

signalled, not kept: a facilitator or verifier rejecting a receipt *“SHOULD signal the reason via the* `X-Receipt-Reject-Reason` *response header”*, and nothing in the draft requires storing the rejected attempt. The one store it does mandate is for replay rather than for refusals — §6.2’s nullifier store must be durable and must reject a repeated nullifier, which records what was spent and not what was turned away. *So the refusal has a code, a closed vocabulary and a transport, and no population.* **A code for the refusal and a record for the success, written by someone who was not making the argument.** It is counted here as an instrument and **not** among §4’s eight: its signalling duty is a SHOULD, where each of the eight requires a record.

MiCA’s record-keeping standard (Delegated Regulation 2025/1140, under Regulation 2023/1114 Art. 68(9)) requires records of client communications *“including where such conversations or communications do not result in the conclusion of a transaction”*, records of clients’ consent, *“all instructions received”*, and *“Records of clients who did not provide the information required by Article 81(8)”* — an intake population with refusals and their reasons included — and for the

beneficiary and intermediary roles the dispositions are three, not two: *"any return, suspension, or rejection of the transfer of crypto-assets"* with *"the reason for such return, suspension, or rejection."* **Scoped to crypto-asset transfer services** — that clause is part of the sentence, not a caveat on it. **And the Annex bounds itself.** Its structured order record is a table of 41 fields, and no field in it requires a reason for an order that was not executed, was cancelled, rejected, suspended or returned: the events and statuses are carried as codes and the transmission field is a boolean. (Described rather than quoted: that section was read in summary, because the instrument capped quoted lines at 125 characters.) *It reaches a kept refusal population, in one sector, with its reasons in prose and its population in codes, and it does not join them.*

The PSD2 strong-customer-authentication standard (Regulation 2018/389 Art. 29) requires traceability enabling *"knowledge ex post of all events relevant to the electronic transaction in all the various stages"* — session identification, detailed logging, transaction numbers, timestamps. *It reaches a duty to be able to reconstruct, without stating a population or a rate.*

Nacha's rules require an originator to retain proof the receiver *"affirmatively initiated each payment in accordance with the terms of the standing authorization."* *It reaches the grant, and the link from an initiation to it.*

The Visa Core Rules (§4.1.23 in the 24 October 2026 public edition, **rule text last updated April 2026** — the agentic rules are not new in the October edition) require cardholder consent, cardholder-defined criteria, retention of the stored-credential agreement for the duration of the agreement on written request from the cardholder or issuer, and — §4.1.23.8 — that *"Following the completion of an Agentic Transaction"* the provider *"must make available to the Cardholder, for at least 120 days from the Processing Date, an order confirmation."* **The operative verb is make available, to the one party that already knows what happened, and the trigger is completion.** §4.1.23.10 places the liability on the cardholder *"as if the Cardholder initiated the Transaction." They reach consent, criteria, and a confirmation of what completed, shown to a party who was there.* Mastercard's Transaction Processing Rules §3.15.2 carries 13-month acquirer retention, also public.

`draft-schrock-action-evidence-boundary` reaches the strongest refusal-recording duty of any instrument here, and it is narrow: a MUST at the pre-entry recheck, a SHOULD at the evaluator, and over the portion read no counter, no sequence number and no denominator anywhere in it. *It reaches a required marker for one class of refusal, bound to the attempt, written atomically with the close.*

The Universal Commerce Protocol's AP2 Mandates extension (2026-08-25) requires signed checkout terms and signed mandate proof at completion once terms are negotiated. *It reaches a signed grant and a signed close, for a completed checkout.*

Uniform Commercial Code Article 4A, whose relevant sections date from 1989, makes the point most plainly. It **names** the refusal, **serves** it, and gives it a terminal effect: §4A-210 is headed *"REJECTION OF PAYMENT ORDER"*, a bank rejects by a notice to the sender orally, electronically or in writing, *"A notice of rejection need not use any particular words"*, and §4A-210(d) makes the states exclusive — *"Acceptance of a payment order precludes a later rejection of the order."* **And it requires no record of any of it.** Over the sections read, the words *record* and *retain* do not appear on the rejection page

at all, and nothing requires a rejection to be retained, logged or counted; §4A-212 says that *"except as provided in this Article or by express agreement"* a receiving bank *"does not otherwise have any duty to accept a payment order or, before acceptance, to take any action"*, and §4A-211(d) cancels an unaccepted order by operation of law after five business days. *It reaches a named refusal with a terminal effect and no record of it.* **And it is the strongest form of the pattern rather than the same form: the eight documents in §4 all REQUIRE the refusal record and leave the population unverifiable, where this statute does not require the record at all.** The 2026 documents are the near end of a gradient, not a departure from it. Scope, because it is narrow and because the claim is a negative: §4A-108 defers to the Electronic Fund Transfer Act, and a payment order is a credit-push instruction to a bank, so the Article reaches wholesale credit transfers and not the card rails most agentic commerce sits on. **And this is a statement about the Article, not about US law.** Fifteen of its thirty-eight sections were read; §4A-501 lets a funds-transfer system's own rules displace the Article even where they conflict with it; and bank recordkeeping law

> is a different body of law that this paper has not opened. **Nothing here says a US bank need not record a rejected payment order** — only that this statute does not require it.

4. Eight authors reached part of this first, the count is the argument, and they are not one community

This argument is not new and should not be presented as new. **Eight documents reach part of it** — Hawkins, Farley, Sharif, Kamimura, Abak, Schrock, **Sato** and **Peng and Wu**. Citing the convergence is a stronger position than claiming the observation, and a paper that presents as novel an argument eight other authors have published is the weaker paper.

But the convergence is not the finding. This is: in all eight, the refusal is required, and **no party who was not present can establish that the record is the whole record.** That is a tabulation of each document's own normative language, checkable without agreeing with the author about anything.

The disposition of that second half differs across the eight, and after the Sato read the paper no longer flattens it. In six it is *optional, unenforced or absent.* In the seventh it is **present, mandatory on the producer, and declared non-conclusive by the draft itself** — which is a stronger instrument and the same result. **The eighth is not counted in this disposition at all**, because it carries no normative markings to count: see the note below the table. *A weaker formulation of this — "optional, unenforced, or absent" of all eight — is false as written of Sato, whose counter duty is a MUST. The claim that survives is the one above.*

And the eighth is not an Internet-Draft, which is the part that matters. A count of six would have been six Internet-Drafts in or around the SCITT community, and a reader would be entitled to answer that a shared drafting culture produces a shared blind spot. *Stateful Governance for Concurrent Agentic Systems* is a systems paper from Purdue and the University of Maryland that arrives at the same shape by a completely different route — **a serialisability argument about concurrent effects, with no transparency log, no receipt format and no SCITT vocabulary in it.** Two literatures that do not cite each other, reaching the same boundary, is not a convention. **That sentence is measured in both directions and the measurement is the whole of its warrant.** Forward: StatefulGovernance contains SCITT 0, IETF 0, RFC 0, `draft-` 0, Internet-Draft 0, transparency log 0, receipt 0, and zero occurrences of any of the six draft authors' names. Reverse, swept over the full text of all six drafts, **and over the reference section of each one, which took three re-reads to reach — Sharif and Schrock on 2026-11-12 and Abak's §§17.1–17.2 on the same day, because the first read of each stopped short of it**: no draft cites Peng and Wu, the arXiv identifier, the title, or any serialisability line — and the one that could most easily have, did not. **Hawkins cites two arXiv preprints and not this one**, so the omission there is a choice rather than a convention against citing preprints. **Sharif and Schrock cite no arXiv paper and no DOI at all**, so in their case it is a convention and the sentence claims nothing more about them than that. **Abak's twenty-six references are Internet-Drafts and project artefacts, with three DOIs that are all RFC DOIs** — no preprint of any kind, which is a third shape again and not evidence either way about this paper.

document	the refusal is required	what would make the population checkable is not
Farley, `acta-signed-receipts-03`, 29 Aug 2026	§3.1.1 *"decision (REQUIRED): The policy evaluation result. One of: 'allow', 'deny', 'rate_limit', 'require_approval'"*	`previousReceiptHash` is OPTIONAL, *"present at most once"*; §9.1 *"Receipts include an issued_at timestamp but do not include a nonce or sequence number"*
Sharif, `agent-audit-trail-09`, 26 Oct 2026	§4.2 requires a denied decision be written **before** execution — *"the record MUST have record_phase set to 'pre_execution'"* — and `prev_hash` is REQUIRED, with §8.2 *"all records MUST form a strictly ordered chain"*	The chain is tamper-evident over what it contains and establishes nothing about what never entered it. `deny_reasons` is OPTIONAL; `policy_digest` is OPTIONAL and is *"The SHA-256 hash (lowercase hex) of the canonical classification policy"* — the ruleset mapping action types to trust levels, not the evaluation that produced the decision
Hawkins, `scitt-attested-agent-payment-01`, 15 Aug 2026	§4 *"An executor MUST distinguish, in whatever it reports to the instructing party"* between *"authorization refused"* and *"authorization could not be evaluated"*, and must record the action taken on an indeterminate result	The refusals are not in the digest. §5.1: *"For each payment settled under a scope, the executor MUST compute an execution digest"* — settled payments only; the disposition in force is a SHOULD; the report goes to the instructing party. **The `-02` revision is a Notice of Discontinuation**
Schrock, `action-evidence-boundary-07`, 25 Sep 2026	§5.10's not-entered marker, a MUST, bound to the attempt, in the same atomic write as the close	The evaluator's record is a SHOULD, and over the portion read there is no counter, no sequence number and no denominator anywhere in it
Kamimura, `scitt-refusal-events-02`	§3.6 makes DENY a first-class outcome inside an explicit identity — *"there MUST exist exactly one Outcome (DENY, GENERATE, or ERROR) with attempt-id equal to A"*, written `|{ATTEMPT}| = |{DENY}| + |{GENERATE}| + |{ERROR}|`	*"Transparency Services do not enforce this invariant"*; registration is a SHOULD; §7.2 *"if an ATTEMPT is never logged, this specification cannot detect the omission"*
Abak, `agent-control-delivery-evidence-02`, 29 Aug 2026	§6.4's seven-term identity over a closed population — *"Delivery attempts bind evidence to members of O and do not add elements to O"* — and §5.11's *"The report MUST publish |I| alongside |O|"*	§5.15: *"A producer or configuration declares the set closed"*, and *"the relying party lacks accepted evidence for the stronger enumeration-completeness claim"*

document	the refusal is required	what would make the population checkable is not
Sato, `sato-soos-mjwt-04`, 13 Aug 2026	**The most explicit refusal apparatus of the eight.** §8.2 is a table of twenty-one deny codes bound to numbered verification steps. §7.4 fails closed *with a recorded disposition*: if the human-escalation timeout expires, the action MUST be treated as DENY and *"The GAR record MUST carry resolution: TIMEOUT_DENY"*, with *"No implicit consent is ever inferred from a failure to escalate."* §7.3 requires a `MANDATE_REVOKED` event per revocation carrying a reason, a revoking principal and a timestamp	**The population is named and still not checkable.** §10: the Session Audit Record *"includes the mandate_id of every MJWT presented during the governed session"* — a stated denominator. Nothing in the profile lets a party who was not present establish that the GAR holds every presentation; the GAR is `draft-sato-soos-gar`, **desk-read in full on 2026-11-12 — and it is the strongest instrument in this table, which is why it stays in it.** Its `session_sequence_number` is *"a monotonically increasing integer assigned by the GEC at session open"* and **the duty on the producer is a MUST** — *"The GEC MUST increment this counter for every governed session and MUST NOT reuse values"* — **in `-08`, dated 2026-09-07. The field and its gap-detection language are present from `-05`, 2026-08-18; the producer-side MUST is read in `-08` and is not established for `-05`.** A missing number therefore means a session existed and is absent. **Three things still stop a party who was not there establishing the set is the whole set.** The check is on the other side and optional — *"Verifiers SHOULD check for gaps in `session_sequence_number` sequences as an indicator of suppressed sessions."* **The draft then says the check is not conclusive, in its own words:** *"A gap alone is not proof of suppression: legitimate concurrent-session counter races can produce apparent gaps that later resolve as later-numbered sessions commit."* And the one place the draft uses the word is about a chain, not a population — *"Causal completeness: The ALE causal chain*

document	the refusal is required	what would make the population checkable is not
		for any session MUST be reconstructible from the ALE store alone” — which is **inclusion, not completeness**: a chain that reconstructs says nothing about the sessions that never entered the store. *Scope:* `-08`*, 165,951 characters, read in full; the concession is* `-08`*’s and the counter is in* `-05`
Peng and Wu, *Stateful Governance for Concurrent Agentic Systems*, arXiv 2608.02764v2, 10 Aug 2026	**Required for the theorem, not by a keyword.** *“A denied operation records a denial and produces no policy-state effect”*, and the serialisability result depends on it: *“because the serial order follows the points at which these effects commit or denials are recorded…”* — a denial is a first-class event in the same history as a commit, by construction. *“audit records must justify delayed or denied operations”*	**No signature anywhere in it.** Over the full text: signature 0, *non-party* 0, *third party* 0, *external verification* 0, merkle 0. The history is the provider’s own transactional ledger, implemented as *“SQL tables, counters, or service-owned logs.”* Nothing is offered to a reader outside the provider

Eight for eight on the requirement. The vocabulary exists; in seven of the eight it carries normative force and in the eighth it is a premise rather than a norm; and in every one of them it stops at the boundary of the recorded set.

Two honest notes on the last two rows, because they are not the same kind of document as the first six. Sato’s is an Internet-Draft like the others and sits in the table on the same terms. **Peng and Wu’s is not normative at all** — it has no MUST to quote, and *“requires”* there means *the correctness argument does not hold without it*. **That is a different kind of requirement — a premise of the serialisability result rather than a normative one — and it is marked here rather than smoothed over.** It is a preprint, not a peer-reviewed publication, and §9 gives the version and the date.

Two of the drafts say so about themselves, in better words than this paper had. Abak §6.4: *“Obligation accounting alone cannot demonstrate that every instruction was considered.”* §5.11: *“Closing a target population for reconciliation is not equivalent to proving complete mediation.”* §9.3: *“A set being closed for a reconciliation run does not, by itself, prove that the enumeration is complete.”* §10.3: *“An open population cannot support a completeness claim.”* And §10.3 names the two attacks: *“A producer can suppress failed controls, omit a required target, or choose a favorable subset after the fact”*, and *“A target set can also be made deceptively small while remaining internally closed.”*

Kamimura calls its own rule a completeness invariant — *“The completeness invariant is the core integrity property”* — which is the word this paper should use, because the draft names as its core property the thing it then says it cannot enforce. The sharper statement is not that the drafts lack a word for completeness. It is that one of them has the word, states the arithmetic, and says in its own §7.2 that it cannot detect the omission.

So the demand has four rungs, not two, and the top one is reachable.

rung	what is demanded	who occupies it
weak	records from which an auditor can calculate a rate	most of §3
middle	a stated conservation identity over a declared population, unenforced	**Abak and Kamimura**, Abak more fully; **Sato's named GAR population** sits here too
strong	proof that the population is the whole population, checkable by someone who was not there	**nobody read here, including Certificate Transparency**
reachable	the party relying on the record refuses anything not in it, so the accepted set and the recorded set coincide	**Certificate Transparency's enforcement loop** — §6

Naming the middle rung locates the gap where it is: enforcement and completeness, not vocabulary and not arithmetic. Naming the fourth stops the paper ending on a rung nothing can stand on.

One scope note on the table's Abak row: the seven-term identity partitions delivery obligations, whose non-success term is EXPLICIT_FAILURE. Abak's REFUSED sits in a separate enforcement dimension. They are treated here as the same kind of thing — a non-success inside a stated denominator — which is a reading and not the draft's own equation. ## 5. What is left, stated as narrowly as the evidence allows

One claim survives, and it is the completeness claim. In none of the instruments read can a party who was not present establish that the set they are looking at is the whole set. Abak says so of itself. Kamimura says so of itself. The regulations require records without requiring that the record be provably exhaustive. *"Requiring cryptographic proof that nothing was omitted is a stronger demand than requiring records from which an auditor can calculate a rate"* — and the stronger demand is unmet in every instrument this paper opened.

A ninth document states the general form of the claim about a different subject, and its sentence is the best one in this paper. `draft-noa-scitt-ai-agent-receipt-01` is reasoning about whether two signatures on a receipt come from independent parties, not about populations at all, and it arrives at: *"A self-declared relationship — a producer writing 'independent' about itself — is not evidence of independence, and an undetermined relationship cannot be upgraded to an independent one."* **Substitute *complete* for *independent* and that is §4's entire tabulation in one line.** The same draft refuses to let a producer-side control stand in for a checkable property — a gate that enforces something at emission *"binds nobody downstream and a verifier MUST NOT assume it ran"* — and states the reason in the form this paper has been circling: *"What it changes is what the issuer knows, not what a later reader can check."*

The refusal-denominator claim narrows, and the narrowing matters. It cannot be said that refusals go uncounted: they are counted, monthly, at network scale, in an explicit ratio with declines in both terms. What cannot be done is check the count. Visa's enumeration ratio is computed from VisaNet, which only Visa holds, and the fact sheet states no verification or dispute right for the acquirer or the merchant — so the rate is an assertion by the counter. Whether Visa's own rule book provides a representment or audit right over those counts is not something this paper has read, and the claim is scoped accordingly. The narrow true statement is: **a**

refusal-inclusive denominator exists; a refusal-inclusive denominator that someone other than its author can verify does not.

And the strong rung has no existence proof anywhere this paper looked, including the place one would expect to find it. Certificate Transparency is the most mature deployed transparency system, and RFC 9162 requires treating each log as a trusted third party and states that its own auditing can be circumvented by a misbehaving log. A reader entitled to think the strong demand utopian is owed that answer rather than a reassurance: on the documents read, it has not been met in this field or in the field with the best machinery for it.

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6. What Certificate Transparency does show, and it is the fourth rung

This is the rung §4's table calls reachable, and CT is the only thing in this paper that occupies any rung above the middle.

CT does not prove its logs complete. What it does is make the *accepted* population and the *logged* population the same population, by a mechanism that sits nowhere near the log: **the relying party refuses to honour an item that is not in it.** RFC 9162's abstract states the intent — *"The intent is that eventually clients would refuse to honor certificates that do not appear in a log"* — and the enforcement that followed is a matter of browser policy rather than of the RFC, which is why this paper cites the intent and not the practice. **That primary is unread here and the claim is held to what the RFC says.**

The distinction is worth the paragraph because it redirects the demand. *Prove the record is complete* is a demand on the keeper of the record, and the keeper is the party whose conduct is being measured — which is why every instrument in §3 ends up with a declared population. *Refuse what is not in the record* is a demand on the reader, and the reader has no interest in a short denominator. The second is a design someone can actually ship; the first has been asked for by eight documents and supplied by none.

What that implies for measurement is narrow and worth saying plainly: **report the population before the rate.** A conformance figure without the population it was computed over is not a weaker version of the same number, it is a different kind of statement. The author's published measure definitions treat coverage as the first measure for that reason, and publish the three uncertainties on one conformance figure so a reader can see which one dominates: unidentified coverage 23.8 points, the coding choice 11.3 points, sampling error 6.5 points on a Wilson 95% interval. **An interval published alone understates what the reader does not know by about 17 points.**

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7. The limits of the thing being argued for

Observation needs no partner. The record does. A profile that only watches makes no outbound call, and that is what removes the largest adoption blocker — but the half of a record that attests a success requires a counterparty, because the reference that proves the movement comes from whoever performed it. Nothing here claims otherwise, and a paper that elided it would be making the easy half of the argument.

A kept refusal population is not a proven one. Everything in §3 that keeps refusals keeps them because a rule says to. A party that does not write the refusal down leaves no trace of not having written it down, which is Abak's §10.3 and Kamimura's §7.2 in two different vocabularies. This paper does not solve that and says so.

Nothing here is a claim about what any party intends. The Visa finding in §2 is about where a denominator sits, not about motive: a programme that counts declines where its own exposure lies and settlements where conduct is assessed is behaving sensibly on its own terms. The observation is that the two shapes exist side by side in one publication, which is what makes it evidence rather than inference.

What this paper withdrew, and what broke each one. Four claims an earlier draft of this argument made are gone. None was dropped on an editor's judgement: each was broken by a constructed attempt to break it, or by a read that contradicted it, and in two cases the attempt ran before the sentence had been published.

"No instrument requires the grant, the close and the join, and no format expresses it." **Withdrawn, not narrowed.** The Agent Payments Protocol's receipt binds to the closed mandate by hash in a required field, and MiCA's record-keeping standard requires an intake population that includes communications concluding no transaction. The join exists, it is cryptographic, and in one sector the population is mandatory.

"No instrument supplies evidence a ledger committed the movement in a form a stranger can read." **Withdrawn, and so is the narrowing that replaced it.** The claim fell to x402's prioritised EIP-3009 path, whose settlement emits a public event carrying an authorisation nonce. The narrowing then offered in its place had been built on the *fallback* path, and a quotation supporting it was found not to exist in the draft it was attributed to.

"`draft-schrock-action-evidence-boundary` *does not require a refusal to be recorded."* **False on its first clause.** The not-entered marker is a MUST, bound to the attempt, in the same atomic write as the close. The second clause — that the evaluator-side recording language is a SHOULD — survives.

Certificate Transparency as the existence proof that the stronger demand has been met in another field. **Backwards.** RFC 9162 states that its logs do not themselves prevent misissuance, that its auditing can be circumvented by a misbehaving log, and that it is therefore necessary to treat each log as a trusted third party. What CT occupies is the fourth rung, and §6 says what that is worth.

Listing them is not an exercise in candour. A sentence that has survived an attempt to break it is worth more than one nobody attacked, and three of the four that replaced these are narrower than what they replaced.

And the scope of every negative here is the set of documents at the foot. Several are read in part — the character ranges are recorded — and a claim about what an instrument does not contain is a claim about the portion read.

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8. What follows

Three things, none of which needs anyone's permission.

Report the population first, and publish how the figure lies. A conformance rate travels with the coverage it was computed over or it does not travel. This is cheap, it is unilateral, and it is the one move that makes the rest legible.

Keep the refusals in the same stream as the successes. Not because a regulator requires it — in one sector one does — but because a denominator assembled later from a different store is a denominator nobody can check, including its owner.

Ask the reader-side question rather than the keeper-side one. The eight documents in §4 all ask the keeper to prove its population. CT suggests the other question: what would a verifier have to refuse for the kept population and the real one to coincide? That is a design question, it is answerable, and nothing in the instruments read has answered it.

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9. Sources

Every instrument quoted above, with the route, the date opened, and the portion actually read. A claim that an instrument does not contain something is a claim about the portion named here and no more.

Instrument	Route	Opened	Portion read
Visa Acquirer Monitoring Program fact sheet 2025	`corporate.visa.com` (Visa's own publication of its programme)	2026-11-11	Both ratio definitions, thresholds, monitoring cadence. **Not** the Visa Core Rules pages
Visa Core Rules §4.1.23, public edition, 24 Oct 2026	Visa public rules	2026-11-10	§4.1.23 and §4.1.23.10
Mastercard Transaction Processing Rules, public edition	Mastercard public rules	2026-11-10	§3.15.2
RFC 9162, Certificate Transparency v2	`rfc-editor.org/rfc/rfc9162.html`	2026-11-11	Abstract and §1 only
AP2 Payment Mandate / Payment Receipt	`ap2-protocol.org/ap2/payment_mandate/`	2026-11-10	Receipt schema and the field-presence rule
x402 exact scheme, EVM	`github.com/coinbase/x402`, `specs/schemes/exact`	2026-11-11	The scheme file and README; both settlement paths
`draft-vauban-x402-consolidated-00`**, 28 May 2026** (F. Serafini, Ed., Vauban Research)	`ietf.org/archive/id/`	**2026-11-12**	**§3.13 Error Taxonomy including Table 5's seven tokens with their HTTP status codes, the closed-set sentence, the `X-Receipt-Reject-Reason` duty, and §6.2's nullifier store. Informational, an Independent Submission, not a Working Group product. It is an extension to x402 V2 and NOT x402 itself: the seven tokens are not in the x402 repository cited**

Instrument	Route	Opened	Portion read
			in the row above, and an earlier draft of this paper attributed them there. §§1–3.12, 4, 5 and 7 onward swept only
Delegated Regulation (EU) 2025/1140 (under Reg. 2023/1114 Art. 68(9))	`eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32025R1140` ; OJ L 2025/1140, 10.6.2025	2026-11-10	**Annex Section 1 near-verbatim. Section 2 in summary only — the instrument capped quoted lines at 125 characters, so nothing from it is quoted here. Section 3 through field 14. Section 4 not read**
Regulation (EU) 2018/389, SCA RTS	EUR-Lex	2026-11-10	Art. 29
`draft-hawkins-scitt-attested-agent-payment-01` , 15 Aug 2026	`ietf.org/archive/id/`	2026-11-11	Full; §§4, 5.1, 6. `-02` is a Notice of Discontinuation
`draft-farley-acta-signed-receipts-03` , 29 Aug 2026	`ietf.org/archive/id/`	2026-11-11	Full; §§2.2, 3.1.1, 9.1
`draft-sharif-agent-audit-trail-09` , 26 Oct 2026	`ietf.org/archive/id/`	2026-11-11; re-opened in full 2026-11-12	§§3.1, 3.2, 4.2, 5.3, 8.2 on the first read (characters 0–100,000 of 108,942). **Re-opened 2026-11-12 through the browser pane, 151,882 characters, reaching §§15.1–15.2 References** — the portion the first read could not: WebFetch truncated at every offset tried, which is an instrument fault and not an absent section
`draft-schrock-action-evidence-boundary-07` , 25 Sep 2026	`ietf.org/archive/id/`	2026-11-11; re-opened in full 2026-11-12	§§5.6, 5.8, 5.10, 5.11 on the first read (characters 0–100,000 of 126,103). **Re-opened 2026-11-12 through the browser pane, 148,442 characters, reaching §§16.1–16.2 References.** The §4 table's

Instrument	Route	Opened	Portion read
			claim about this draft is still the first read’s scope: no counter, no sequence number, no denominator over §§5.6–5.11
`draft-abak-agent-control-delivery-evidence-02` , 29 Aug 2026	`ietf.org/archive/id/`	2026-11-11	Characters 0–100,000 of 123,860; §§4.2, 5.10, 5.11, 5.15, 6.4, 9.3, 10.3. Conformance cases unread
`draft-kamimura-scitt-refusal-events-02`	`ietf.org/archive/id/`	2026-11-11	Full; §§1.3, 3.6, 5.2, 5.3, 7.2
`draft-sato-soos-mjwt-04` **, 13 Aug 2026**	`datatracker.ietf.org`	**2026-11-12**	**Full text, 89,953 characters after whitespace collapse; §§5, 7.1–7.5, 8.1, 8.2 and 10 read verbatim, the rest swept. This is** `-04` **of** seven**: the family runs** `-00` **to** `-06` **and** `-06` is dated 2026-09-07**, read off the datatracker document page 2026-11-12.** Two revisions after the one quoted are unopened**, so every quotation here is scoped to** `-04` **and nothing is claimed about the current text. No material marked *[NEW in -02]* is quoted in this paper**
`draft-sato-soos-gar` **,** `-00` **17 May 2026 through** `-08` **7 Sep 2026**	`ietf.org/archive/id/` **, browser pane**	**2026-11-12**	`-08` **read in full, 165,951 characters;** `-00` **,** `-04` **and** `-05` **opened on their own headers for revision bracketing, so which revision introduces which property is bracketed rather than pinned.** `-00` **2026-05-17: no counter, no Merkle, no SCITT.** `-04` **2026-08-13: the Merkle and SCITT vocabulary.** `-05` **2026-**

Instrument	Route	Opened	Portion read
			08-18: `session_sequence_number` and the gap-detection language. `-08` 2026-09-07: the producer-side MUST and the not-proof concession — so every quotation in §4 is attributed to the revision carrying it. `-01`, `-02`, `-03`, `-06` and `-07` are unopened
Peng and Wu, *Stateful Governance for Concurrent Agentic Systems*, arXiv 2608.02764v2, 10 Aug 2026 (v1 submitted 2026-08-03)	**arXiv `2608.02764`, version 2 of 2. A preprint — not peer-reviewed, not a conference or journal publication, and not an Internet-Draft. Authors at Purdue and the University of Maryland. *MasuGate* is the system name, given in parentheses beside the title, not an alternative title**	**2026-11-12**	**Full text, 93,057 characters; the deny, denial, record, audit, spent, ledger and policy-version families read in context, and probed separately for signatures and external verifiability. Quoted from v2 throughout**
`draft-noa-scitt-ai-agent-receipt-01`	**`ietf.org/archive/id/`, browser pane**	**2026-11-12**	**Full text, 200,367 characters; §§4.2 and 6.1 read verbatim. Quoted in §5 only, and NOT in §4's tabulation — the sentences quoted are in the part of §6.1 the draft itself marks NON-NORMATIVE**
`draft-toraman-noa-action-digest-01`, 14 Aug 2026 (M. Toraman)	`ietf.org/archive/id/` and the `ftp.kaist.ac.kr` mirror	**2026-11-13, DESK**	**Re-read in full, 45,241 characters, through the browser pane, after the §6.2 quotation an earlier revision carried was found not to exist. §5 (modes (a) and (b), E1, E2, E3) and §6 (all six steps and the closing paragraphs) verbatim; §§1–4 and 7 swept. THERE IS NO §6.2: `6.2` ×0, `exactly one` ×0, `(authorizer` ×0. Confirmed on both**

Instrument	Route	Opened	Portion read
			publishers and across `-00` and `-01`. A WebFetch relay capped every quotation at 125 characters and reported one fragment as exact that the full text does not contain, which is why this row is a browser-pane read
`draft-das-payment-execution-finality-01`, (A. Das)	the `ftp.kaist.ac.kr` mirror	**2026-11-13, DESK**	**Opened to test whether §4's count is eight or nine, 87,301 characters. §10 step 5 reads *"On fail: deny; … commit denial evidence (a denial LAVR or equivalent protected enforcement record) as evidence of non-completion. Do not post."* — but it sits in an End-to-End Workflow walkthrough in the imperative, with no RFC 2119 keyword. The draft HAS 2119 boilerplate and uses MUST 42 times, SHOULD 17, MAY 13, so it knows how to require this and does not: `MUST record` ×0 and `MUST commit` ×0 across the whole draft. `denominator` ×0. NOT counted among §4's eight, on the same normative test the eight are counted by**
Nacha Operating Rules, standing-authorisation provision	secondary source	—	**Not read on the rule text**
UCC Article 4A, 2012 uniform text	`law.cornell.edu/ucc/4A/`, browser pane	2026-11-11	§§4A-103, -104, -108, -201, -202, -203, -204, -205, -210, -211, -212, -501 in full and three in part — **15 of 38 sections, read by a second desk,**

Instrument	Route	Opened	Portion read
			not this one. §§4A-210 and 4A-211 independently confirmed here on a different route. **Official Comments not reached on either route.** The 2022 UCC Amendments change §§4A-201 and 4A-202(b)(ii); the text quoted above is the 2012 one
UCP AP2 Mandates extension, 2026-08-25	protocol documentation	2026-11-10	Signed terms and signed mandate proof at completion
The three uncertainties on one conformance figure (§6)	**the author's own measure definitions, Version 2, and the fixture they were computed over**	**measured, not cited**	**Computed rather than sourced, and marked so because every other row in this table is a document somebody else wrote: unidentified coverage 23.8 points, the coding choice 11.3 points, sampling error 6.5 points on a Wilson 95% interval, and the roughly 17-point difference between the first and the last. Definitions Version 2 governs the wording of the measures; the arithmetic is this paper's own, and a reader checking it needs the fixture rather than the definitions**

One row above is not a primary read and is marked so in place: the Nacha provision, which was never opened on the rule text. Two rows are partial and say which part: MiCA's Annex Section 2 and the three Internet-Drafts with character ranges.

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